RESOLUTION MEPC.338(76)
(adopted on 17 June 2021)

2021 GUIDELINES ON THE OPERATIONAL CARBON INTENSITY REDUCTION FACTORS RELATIVE TO REFERENCE LINES (CII REDUCTION FACTORS GUIDELINES, G3)

THE MARINE ENVIRONMENT PROTECTION COMMITTEE,

RECALLING Article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Environment Protection Committee conferred upon it by international conventions for the prevention and control of marine pollution from ships,

NOTING that it adopted, by resolution MEPC.328(76), the 2021 revised MARPOL Annex VI, which is expected to enter into force on 1 November 2022 upon its deemed acceptance on 1 May 2022,

NOTING IN PARTICULAR that the 2021 revised MARPOL Annex VI contains amendments concerning mandatory goal -based technical and operational measures to reduce carbon intensity of international shipping,

NOTING IN PARTICULAR that the 2021 revised MARPOL Annex VI contains amendments concerning mandatory goal -based technical and operational measures to reduce carbon intensity of international shipping,

NOTING FURTHER that regulation 28.4 of MARPOL Annex VI requires reduction factors to be established for each ship type to which regulation 28 is applicable,

HAVING CONSIDERED, at its seventy-sixth session, draft 2021 Guidelines on the operational carbon intensity reduction factors relative to reference lines (CII reduction factors guidelines, G3),

1 ADOPTS the 2021 Guidelines on the operational c arbon intensity reduction factors relative to reference lines (CII reduction factors guidelines, G3), as set out in the annex to the present resolution;

2 INVITES Administrations to take the annexed Guidelines into account when developing and enacting national laws which give force to and implement requirements set forth in regulation 28.4 of MARPOL Annex VI;

3 REQUESTS the Parties to MARPOL Annex VI and other Member Governments to bring the annexed Guidelines to the attention of masters, seafarers, shipowners, ship operators and any other interested parties;

regulation 28.4 of MARPOL Annex VI;

3 REQUESTS the Parties to MARPOL Annex VI and other Member Governments to bring the annexed Guidelines to the attention of masters, seafarers, shipowners, ship operators and any other interested parties;

4 AGREES to keep the Guidelines under review in light of experience gained with their implementation and in light of the review of CII regulations to be completed by the Organization by 1 January 2026 as identified in regulation 28.11 of MARPOL Annex VI, and that annual reduction rates for the period 2027-2030 will be further strengthened and developed taking into account that review.

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ANNEX

2021 GUIDELINES ON THE OPERATIONAL CARBON INTENSITY REDUCTION FACTORS RELATIVE TO REFERENCE LINES (CII REDUCTION FACTORS GUIDELINES, G3)

1 Introduction

1.1 These Guidelines provide the methods to determine the annual operational carbon

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1 Introduction

1.1 These Guidelines provide the methods to determine the annual operational carbon intensity reduction factors and their concrete values from year 2023 to 2030, as referred to in regulation 28 of MARPOL Annex VI.

1.2 The annual operational carbon intensity reduction factors apply to each ship type to which regulation 28 of MARPOL Annex VI applies, in a transparent and robust manner, based on the specific carbon intensity indicators stipulated in the 2021 Guidelines on operational carbon intensity indicators and the calculation methods (G1) (resolution MEPC.336(76)) and the reference lines developed in the 2021 Guidelines on the reference lines for use with operational carbon intensity indicators (G2)(resolution MEPC.337(76)).

1.3 The reduction factors have been set at the levels to ensure that, in combination with other relevant requirements of MARPOL Annex VI, the reduction in CO 2 emissions per transport work by at least 40% by 2030, compared to 2008, can be achieved as an average

other relevant requirements of MARPOL Annex VI, the reduction in CO 2 emissions per transport work by at least 40% by 2030, compared to 2008, can be achieved as an average across international shipping.

1.4 Section 5 of these Guidelines provides background information on rational ranges of reduction factors of ship types in year 2030 using demand- based measurement and supply-based measurement.

1.5 The Organization should continue to monitor development in annual carbon intensity improvement using both demand- based measurement and supply -based measurement in parallel to the annual analysis of the fuel consumption data reported to the IMO DCS.

2 Definitions

2.1 MARPOL means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocols of 1978 and 1997 relating thereto, as amended.

2.2 IMO DCS means the data collection system for fuel oil consumption of ships referred to in regulation 27 and related provisions of MARPOL Annex VI.

2.3 For the purpose of these Guidelines, the definitions in MARPOL Annex VI, as amended, apply.

to in regulation 27 and related provisions of MARPOL Annex VI.

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2.4 The annual operational carbon intensity reduction factor, generally denoted as Z in regulation 28 of MARPOL Annex VI, is a positive value, stipulating the percentage points of the required annual operational carbon intensity indicator of a ship for a given year lower than the reference value.

3 Method to determine the annual reduction factor of ship types

3.1 Operational carbon intensity of international shipping

Given significant heterogeneity across ship types, the attained annual operational CII of international shipping as a whole is calculated as the ratio of the aggregated mass (in grams) of CO2 (aggregated M) emitted to the aggregated mass (in tonnenmiles) of transport work

international shipping as a whole is calculated as the ratio of the aggregated mass (in grams) of CO2 (aggregated M) emitted to the aggregated mass (in tonnenmiles) of transport work (aggregated W) undertaken by all individual ships of representative ship types in a given calendar year, as follows:

/ shippingattained CII aggregated M aggregated W= (1)

In the absence of the data on actual annual transport work of individual ships, the aggregated transport work obtained from other reliable sources, such as UNCTAD, can be taken as approximation. The representative ship types refer to bulk carriers, gas carriers, tankers, container ships, general cargo ships, refrigerated cargo carrier and LNG carriers, as per the Fourth IMO GHG Study 2020.

3.2 The achieved carbon intensity reduction in international shipping

For a given year y , the achieved carbon intensity reduction in international shipping relative to the reference year
refy , denoted as ,shipping yR , can be calculated as follows:

, ,, ,=100% ( ) / ref refshipping y shipping y shipping y shipping yR attained CII attained CII attained CII× (2)

to the reference year
refy , denoted as ,shipping yR , can be calculated as follows:

, ,, ,=100% ( ) / ref refshipping y shipping y shipping y shipping yR attained CII attained CII attained CII× (2)

where the ,shipping yattained CII and , refshipping yattained CII represents the attained annual operational carbon intensity of international shipping in year y and in the reference year
refy , as defined in Eq.(1).

The achieved carbon intensity reduction in international shipping can be alternatively calculated on the carbon intensity performance of ship types. Since CII metrics for different ship types may not be identical, the weighted average of the carbon intensity reduction achieved by ship types can be applied, as follows:

, ,,shipping y type y type y
type
R fR = (3)

In Eq(3),

 type represents the ship type;
 ,type yf is the weight, which is equal to the proportion of CO 2 emitted by the ship type to the total CO2 emissions of international shipping in year y ; and
 ,type yR represents the carbon intensity reduction achieved by the ship type in

,type yf is the weight, which is equal to the proportion of CO 2 emitted by the ship type to the total CO2 emissions of international shipping in year y ; and

,type yR represents the carbon intensity reduction achieved by the ship type in year y , calculated as , ,, ,=100% ( ) / ref reftype y type y type y type yR attained CII attained CII attained CII× ,

where the ,type yattained CII and ,type refattained CII represents the attained annual operational carbon intensity of the ship type in year y and in the reference year refy , as defined in Eq.(4), as follows:

,, / type ship t ship t ship shi p attained CII M W= (4)

where:

,ship tM and ,ship tW represents the total mass of CO 2 emitted from and the total transport work undertaken by a ship of this type in a given calendar year, as stipulated in the Guidelines on operational carbon intensity indicators and the calculation methods (G1).

transport work undertaken by a ship of this type in a given calendar year, as stipulated in the Guidelines on operational carbon intensity indicators and the calculation methods (G1).

4 The reduction factors for the required annual operational CII of ship types

4.1 In accordance with regulation 28 of MARPOL Annex VI, the required annual operational CII for a ship is calculated as follows:

Required annual operatio (1 /100)nal RZ C III CI= ×

where RCII is the reference value in year 2019 as defined in the Guidelines on the reference lines for use with operational carbon intensity indicators (G2) , Z is a general reference to the reduction factors for the required annual operational CII of ship types from year 2023 to 2030, as specified in table 1.

Table 1: Reduction factor (Z%) for the CII relative to the 2019 reference line

Year	Reduction factor relative to 2019
2023	5%*
2024	7%
2025	9%
2026	11%
2027	- **
2028	- **
2029	- **
2030	- **

Note:

* Z factors of 1%, 2% and 3% are set for the years of 2020 to 2022, similar as business as usual until entry into force of the measure.

2023 5%*
2024 7%
2025 9%
2026 11%
2027 - **
2028 - **
2029 - **
2030 - **

Note:

* Z factors of 1%, 2% and 3% are set for the years of 2020 to 2022, similar as business as usual until entry into force of the measure.
** Z factors for the years of 2027 to 2030 to be further strengthened and developed taking into account the review of the short-term measure.

5 Background information on rational ranges of reduction factors of ship types in year 2030

5.1 In the Initial IMO Strategy on Reduction of GHG Emissions from Ships (Resolution MEPC.304(72)), the levels of ambition on carbon intensity of international shipping have been set taking year 2008 as reference. The carbon intensity of international shipping in year 2008, as well as the improvement through 2012 to 2018, has been estimated in the Fourth IMO GHG Study 2020. However, since the scope and data collection methods applied in the Fourth IMO

as well as the improvement through 2012 to 2018, has been estimated in the Fourth IMO GHG Study 2020. However, since the scope and data collection methods applied in the Fourth IMO GHG Study 2020 were inconsistent with those under IMO DCS, the results derived from the two sources cannot be compared directly.

5.2 To ensure the comparability of the attained carbon intensity of international shipping through year 2023 to 2030 with the reference line, the following methods are applied to calculate the equivalent carbon intensity target in year 2030 (,2030shippingeR), taking year 2019 as reference, i.e. how much additional improvement is needed by 2030 from the 2019 performance level.

5.3 The achieved carbon intensity reduction of international shipping in year 2019 relative to year 2008 (,2019shippingR) can be estimated as the sum of the achieved carbon intensity reduction of international shipping in year 2018 relative to year 2008 (,2018shippingR) as given by the Fourth IMO GHG Study 2020 and the estimated average annual improvement during 2012 and 2018 (shippingr), as follows:

,2019 ,2018=shipping shipping shippingRR r + (5)

the Fourth IMO GHG Study 2020 and the estimated average annual improvement during 2012 and 2018 (shippingr), as follows:

,2019 ,2018=shipping shipping shippingRR r + (5)

5.4 The following provides the calculations using demand- based measurement and supply-based measurement.

5.4.1 Demand-based measurement of 2030 target

As estimated by the Fourth IMO GHG Study 2020, the attained CII of international shipping (on aggregated demand-based metric) has reduced by 31.8% (,2018 =31.8%shippingR) compared to 2008, with an estimated average annual improvement at 1.5 percentage points (=1.5%shippingr). In accordance with Eq.(5), the carbon intensity reduction achieved in year 2019 is estimated as 33.3% (,2019 =33.3%shippingR).

5.4.2 Supply-based measurement of 2030 target

As estimated by the Fourth IMO GHG Study 2020, the attained CII of international shipping (on aggregated supply -based metric) has reduced by 22.0% (,2018 =22.0%shippingR) compared to 2008, with an estimated average annual improvement at 1.6 percentage points (=1.6%shippingr). In accordance with Eq.(5), the carbon intensity reduction achieved in year 2019

to 2008, with an estimated average annual improvement at 1.6 percentage points (=1.6%shippingr). In accordance with Eq.(5), the carbon intensity reduction achieved in year 2019 relative to 2008 is estimated as 23.6% (,2019 =23.6%shippingR).

5.5 Given the achieved carbon intensity reduction of international shipping in year 2019 relative to year 2008, the carbon intensity reduction target of international shipping in year 2030 can be converted to the equivalent target (,2030shippingeR) relative to year 2019, as follows:

,2019
,2030
,2019
40%
1
shipping
shipping
shipping
ReR R= (6)
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5.5.1 Demand-based measurement of 2030 target

In accordance with Eq.(6), the equivalent reduction factor of international shipping in year 2030 relative to year 2019 (,2030shippingeR) would be at least 10.0% measured in aggregated

5.5.1 Demand-based measurement of 2030 target

In accordance with Eq.(6), the equivalent reduction factor of international shipping in year 2030 relative to year 2019 (,2030shippingeR) would be at least 10.0% measured in aggregated demand-based CII metric, i.e. at least additional 10.0% improvement from the 2019 level is needed by 2030.

5.5.2 Supply-based measurement of 2030 target

In accordance with Eq.(6), the equivalent reduction factor of international shipping in 2030 relative to year 2019 (,2030shippingeR) would be at least 21.5% , measured in aggregated supply-based CII metric i.e. at least additional 21.5% improvement from the 2019 level is needed by 2030.

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